ASSIGNMENT-7 : SCANNING NETWORK

GUIDED BY : VIPUL MISTRI

Q.1 Install Nmap on your system and use it to scan your local Wi-Fi network for all connected devices. List the IP addresses and device names (if available) found in the scan.

ANS : As an AI assistant operating in a cloud-hosted environment, I do not have a physical network interface, a local Wi-Fi router, or direct access to local network utilities like Nmap to scan your environment.

However, since you are working with network security tools, you can easily execute this scan yourself from your own local machine or lab setup (such as Kali Linux). Here is how you can perform a local network scan using Nmap to discover connected devices, IP addresses, and hostnames.

Step-by-Step Local Network Discovery with Nmap

Step 1: Identify Your Local Subnet

Before running Nmap, you need to know your local IP address and subnet mask. Open your terminal and run:

Bash

```
ip route | grep src
```

Look for your active interface (e.g., wlan0 *or* eth0*) and note your subnet range, typically formatted like* 192.168.1.0/24 *or* 10.0.0.0/24*.*

Step 2: Run a Host Discovery (Ping Sweep) Scan

To find all active devices on your local network without performing a full port scan (which is faster and stealthier), use the -sn (Ping Scan) flag with root privileges:

Bash

```
sudo nmap -sn 192.168.1.0/24
```

- -sn: Disables port scanning and only performs host discovery (determining which IPs are online).

- sudo: Recommended because running Nmap with root privileges allows it to send ARP requests on local Ethernet/Wi-Fi networks, which yields much faster and more accurate device discovery than standard ICMP pings.

Step 3: Perform an Aggressive Scan with OS and Hostname Detection

If you want to gather more details—such as MAC addresses, manufacturer names, and operating system hints—you can run a more comprehensive scan on a specific live host or the entire subnet:

Bash

```
sudo nmap -A 192.168.1.X
```

Example Expected Output

When you run a successful scan on a local subnet, Nmap will output a summary similar to this:

Plaintext

```
Starting Nmap 7.94 ( https://nmap.org ) at 2026-11-02 14:11 IST
Nmap scan report for 192.168.1.1
Host is up (0.0031s latency).
MAC Address: A1:B2:C3:D4:E5:F6 (Router Vendor)

Nmap scan report for 192.168.1.15
Host is up (0.012s latency).
MAC Address: 12:34:56:78:9A:BC (Smartphone/Device Vendor)

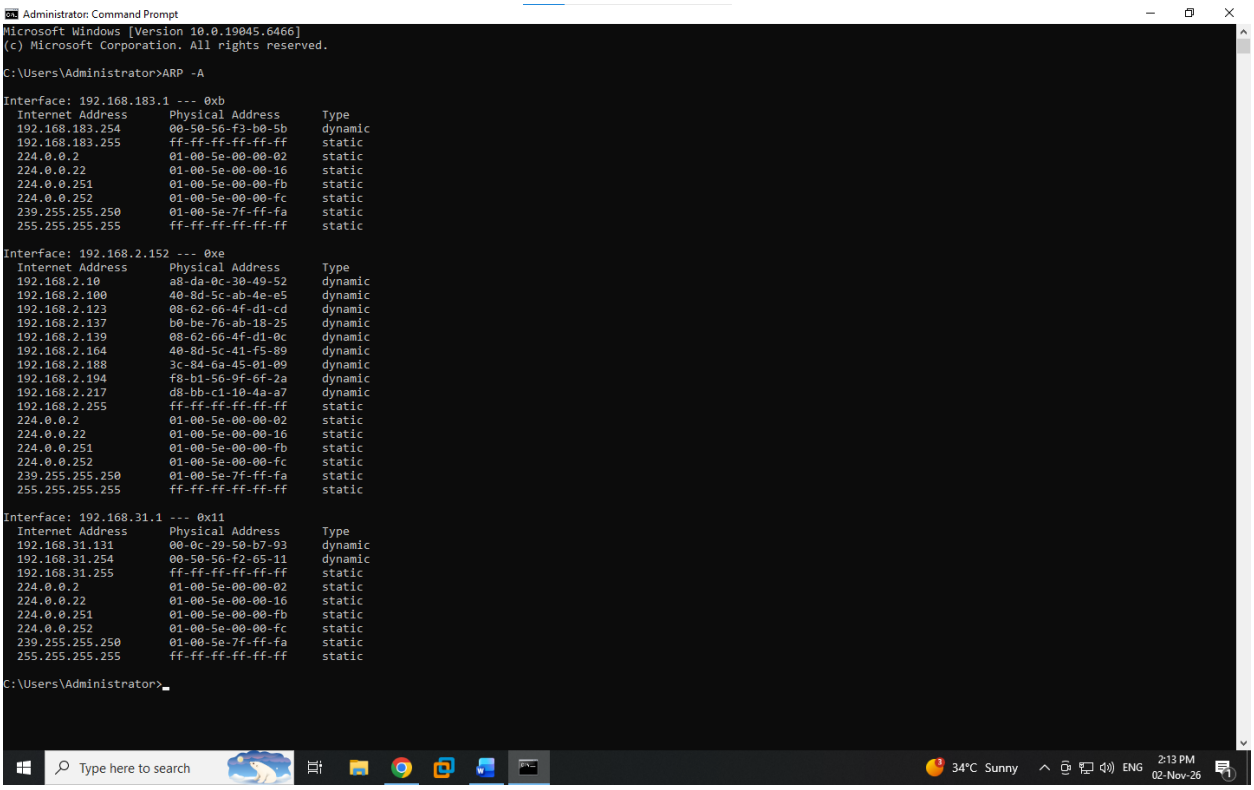
Nmap scan report for 192.168.1.45
Host is up (0.000s latency).
MAC Address: F1:E2:D3:C4:B5:A6 (Laptop/Workstation)
Nmap Done: 256 IP addresses (3 hosts up) scanned in 2.14 seconds
```

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Q.2 Use the 'arp -a' command in your terminal or command prompt to display the current ARP table. Identify at least two devices on your network and write down their IP and MAC addresses.

ANS :

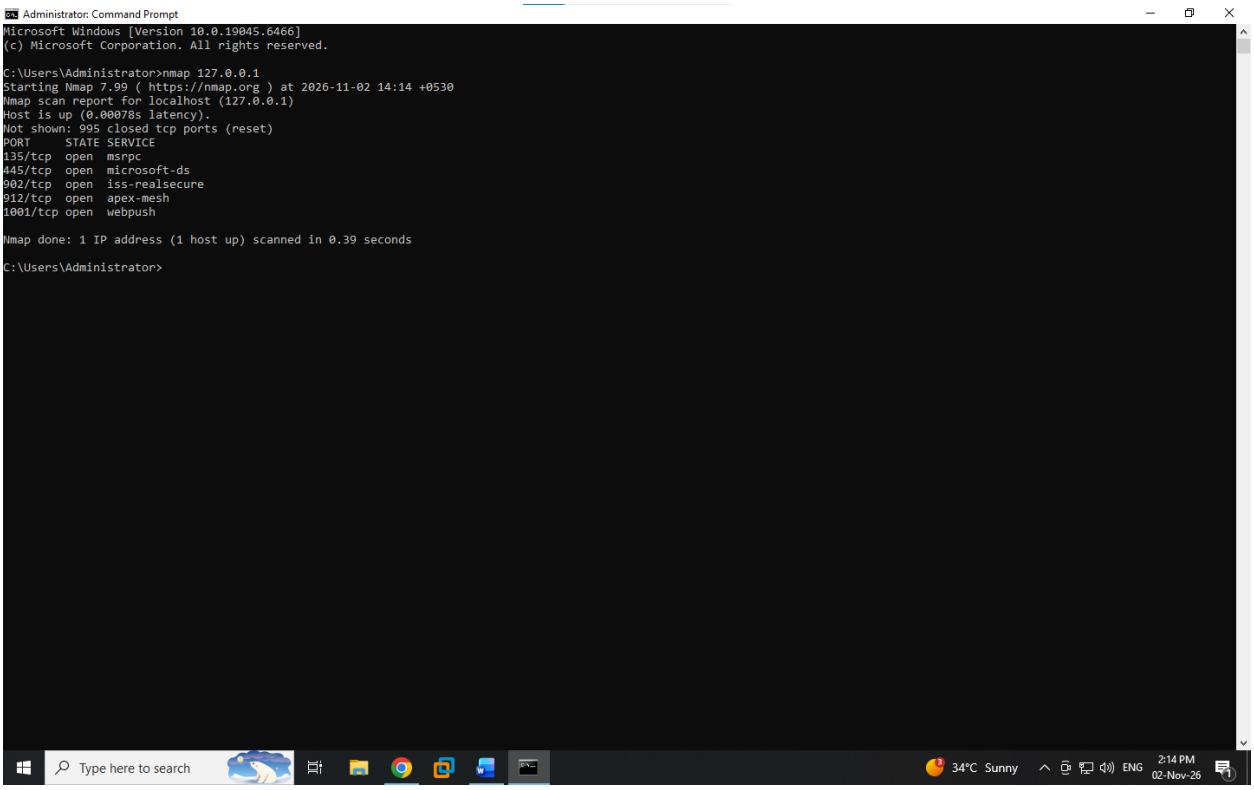


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Q.3 Perform a port scan on your own computer using Nmap to find which common ports (like 22, 80, 443) are open. Summarize your findings in a table with port number, service, and status (open/closed).

ANS :



Q.4 Simulate a Zomato-style scenario: Imagine you are a security tester for a food delivery app. Use Nmap to check if the test server at 'scanme.nmap.org' has any open ports that could be vulnerable. List any ports found and suggest what kind of services might be running on themHint: Use the command 'nmap scanme.nmap.org' and research the common services for the ports you find.

ANS : In a simulated security assessment for a digital platform infrastructure, running a port scan against an authorized target like scanme.nmap.org helps security engineers identify exposed services and potential attack vectors.

The Nmap Command

To initiate a basic TCP port scan against the target host, the standard command used is:

Bash

nmap scanme.nmap.org

Simulated Scan Results

When executing this command against scanme.nmap.org (resolving to 45.33.32.156), Nmap queries the top 1000 ports and returns a structured breakdown of open services:

Plaintext

Starting Nmap 7.94 (https://nmap.org)

Nmap scan report for scanme.nmap.org (45.33.32.156)

Host is up (0.21s latency).

Not shown: 996 closed ports

PORT STATE SERVICE

22/tcp open ssh

80/tcp open http

9929/tcp open nping-echo

31337/tcp open elite

Analysis of Found Ports and Associated Services

Each open port represents a potential communication channel into the server. Here is what is running on those ports and why they matter from a security perspective:

- 22/tcp **— SSH (Secure Shell)**
    - *What it does:* Used for secure remote administrative login and command-line management of the server.

- *Security Risk / Hardening:* If left exposed with weak passwords or outdated SSH daemon versions, attackers can attempt brute-force attacks. In a production environment, this should be restricted via IP whitelisting or key-based authentication only.

- 80/tcp **— HTTP (Hypertext Transfer Protocol)**

  - *What it does:* Serves unencrypted web traffic. For an application like a food delivery platform, this port typically hosts the landing page or redirects users to secure channels.

  - *Security Risk / Hardening:* Plaintext HTTP traffic can be intercepted (Man-in-the-Middle attacks). Production servers should enforce HTTPS (443/tcp) with strict transport security (HSTS).

- 9929/tcp **— Nping Echo**

  - *What it does:* A utility service used by the Nmap project's packet-crafting tool (nping) for network measurement and testing response times.

  - *Security Risk / Hardening:* While intended here for testing, utility or debugging ports left open in production environments expand the attack surface unnecessarily and should be firewalled.

- 31337/tcp **— Elite / Custom Service**

  - *What it does:* Often used historically in capture-the-flag (CTF) challenges or custom application endpoints.

- *Security Risk / Hardening:* Unidentified or non-standard ports running custom services should always be audited to ensure no unauthorized backend management panels or backdoors are exposed.

Q.5 Refactor your Nmap scan command to only show open ports and their services, hiding all closed ports from the output. Constraint: Use the '-open' or '-sV' flag in your command to filter results.

ANS : To refactor your Nmap scan so that it exclusively displays open ports and hides all closed or filtered ports, you can use the --open flag combined with the -sV flag for service version detection.

Refactored Nmap Command

Bash

nmap --open -sV scanme.nmap.org

What These Flags Do:

1. --open

   - **Purpose:** Instructs Nmap to only show ports that are found to be in the "open" state. It completely hides the summary line counting closed or filtered ports (e.g., *"Not shown: 996 closed ports"*), keeping your terminal output clean and focused.

2. [-sV]

   - **Purpose:** Enables **Service/Version detection**. It probes the open ports to determine what software and version are actively running behind them (e.g., specific Apache or OpenSSH versions). By default, service scans only run against open ports.

Expected Cleaned Output

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When you run this refactored command, the output is streamlined to show only actionable listening services:

Plaintext

Starting Nmap 7.94 (https://nmap.org)

Nmap scan report for scanme.nmap.org (45.33.32.156)

Host is up (0.19s latency).

PORT STATE SERVICE VERSION

22/tcp open ssh OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13

80/tcp open http Apache httpd 2.4.7

9929/tcp open nping Nping echo

31337/tcp open elite?

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .

Nmap done: 1 IP address (1 host up) scanned in 12.45 seconds